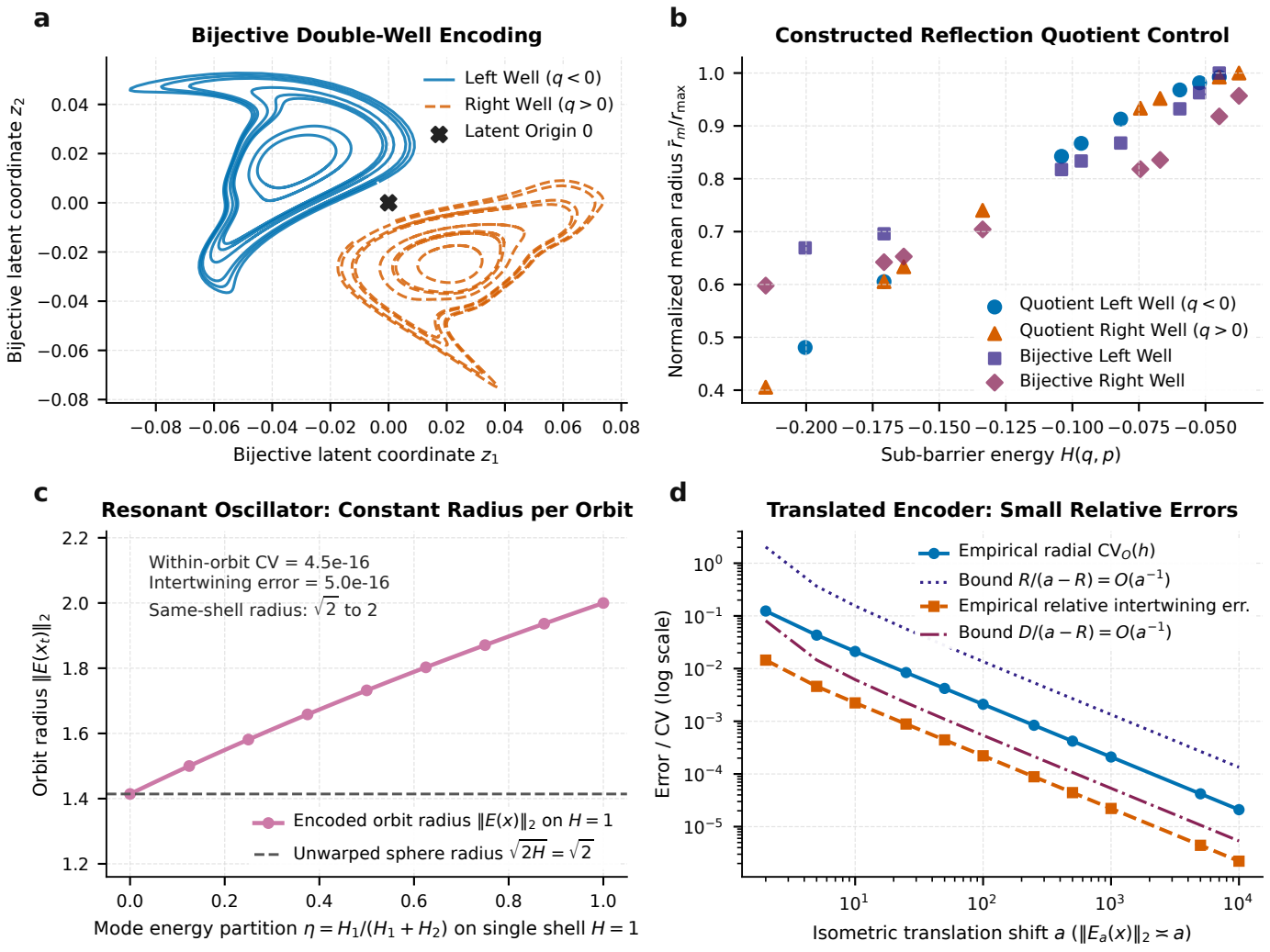

a
Bijective Double-Well Encoding
Left Well ($q < 0$)
Right Well ($q > 0$)
Latent Origin 0
Bijective latent coordinate z_2
Bijective latent coordinate z_1
b
Constructed Reflection Quotient Control
Normalized mean radius $\bar{r}_m/r_{\max}$
Quotient Left Well ($q < 0$)
Quotient Right Well ($q > 0$)
Bijective Left Well
Bijective Right Well
Sub-barrier energy $H(q, p)$
c
Resonant Oscillator: Constant Radius per Orbit
Within-orbit CV = 4.5e-16
Intertwining error = 5.0e-16
Same-shell radius: $\sqrt{2}$ to 2
Orbit radius $\|E(x_t)\|_2$
Encoded orbit radius $\|E(x)\|_2$ on $H = 1$
Unwarped sphere radius $\sqrt{2H} = \sqrt{2}$
Mode energy partition $\eta = H_1/(H_1 + H_2)$ on single shell $H = 1$
d
Translated Encoder: Small Relative Errors
Empirical radial $\mathrm{CV}_O(h)$
Bound $R/(a - R) = O(a^{-1})$
Empirical relative intertwining err.
Bound $D/(a - R) = O(a^{-1})$
Error / CV (log scale)
Isometric translation shift a ($\|E_a(x)\|_2 \asymp a$)